

MATH 147 - Fall 2026

Assignment 1

(due Thursday, September 24 at 5 pm ET on Crowdmark)

Disclaimer: you might see notes in blue/red in the assignments throughout the term. Usually, these include either hints, short explanations, or some motivation for the question, etc.

Problem 1. (10 points)

(i) Prove that $\sqrt{3}$ is irrational.

Why doesn't similar reasoning work for $\sqrt{4}$?

(ii) Using only the ordered field properties of $\mathbb{R}$, together with the fact that for any $c > 0$ there is a positive $x \in \mathbb{R}$ such that $x^2 = c$, prove the arithmetic-geometric mean inequality:

$$\sqrt{ab} \leq \frac{a+b}{2}$$

for all $a > 0, b > 0$ in $\mathbb{R}$.

Problem 2. (15 points)

So far we assumed that the set $\mathbb{N}$ is obviously contained in $\mathbb{R}$. After all, 1 is a real number (it's in the axioms), 2 is just $1 + 1$ and so real, 3 is $(1 + 1) + 1$, etc. In this exercise, you will show that there is a rigorous mathematical way to say "etc.", and define $\mathbb{N}$ as a subset of $\mathbb{R}$.

A set $S \subseteq \mathbb{R}$ is called *inductive* if $1 \in S$ and $k \in S$ implies $k + 1 \in S$.

(i) Determine whether each of the following sets is inductive. Justify your answers.

$$\mathbb{R}, \quad (0, \infty), \quad (0, \infty) \setminus \{5\}.$$

(ii) Show that if S and T are inductive, then the set U of real numbers that are in both S and T is also inductive.

(iii) We define $\mathbb{N}$ to be the set of real numbers that are in every inductive subset of $\mathbb{R}$. Prove that $\mathbb{N}$ is inductive, that is, prove that 1 is a natural number and prove that $k + 1$ is a natural number if k is a natural number.

Problem 3. (20 points) The Principle of Strong (or Complete) Induction states the following. Let $P(n)$ be a sequence of statements with $n \in \mathbb{N}$. If

(a) $P(1)$ is true, and

(b) for every $n \in \mathbb{N}$, if all $P(1), \dots, P(n)$ are true, then $P(n + 1)$ is true,

then $P(n)$ is true for all $n \in \mathbb{N}$.

It might look like the principle of strong induction is much stronger than the ordinary principle of induction, but they are actually equivalent.

(i) Use the Principle of Mathematical Induction to prove the Principle of Strong Induction.

(ii) Let $\{p_n(x)\}_{n \in \mathbb{N} \cup \{0\}}$ be defined recursively by

$$p_0(x) = 1, \quad p_1(x) = x, \quad \text{and} \quad p_{n+1}(x) = xp_n(x) - p_{n-1}(x) \quad (n \geq 1).$$

Prove that for each $n \in \mathbb{N} \cup \{0\}$, $p_n(x)$ is a degree n polynomial and the coefficient of x^n in $p_n(x)$ is 1.

(iii) Let $0 < |t| < 1$. Prove that

$$p_n(t + t^{-1}) = \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} \quad \forall n \in \mathbb{N} \cup \{0\}.$$

(iv) Recall that a (*real*) *root* of a polynomial $p(x)$ is a number $c \in \mathbb{R}$ such that $p(c) = 0$. Use the formula obtained in (iii) to prove that all the real roots of $p_n(x)$ (if any) must be contained in the interval $[-2, 2]$.

Hint: show that the range of the function $g(t) = t + t^{-1}$ defined on $\{t \in \mathbb{R} \mid 0 < |t| < 1\}$ is exactly $\mathbb{R} \setminus [-2, 2]$. Then use the formula in (iii) to show that $p_n(t + t^{-1}) \neq 0$ for all such values of t .

Problem 4. (20 points)

(i) Suppose that it is true that for each $x > 0$ there is an $n \in \mathbb{N}$ so that $\frac{1}{n} < x$. Prove the Archimedean Property using this assumption.

Together with the result proved in class, part (i) shows that the Archimedean Property is equivalent to the statement that for every $x > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$.

(ii) The following statement follows from the Archimedean property (whose proof uses the completeness axiom)

$$\text{if } x > 0, \text{ then there is a natural number } N \text{ so that } \frac{1}{N} < x.$$

Give a proof that does not use the completeness axiom that works when x is rational. Then find a proof that is valid for $x = \sqrt{y}$ where $y > 0$ is rational.

This exercise says that although the result for arbitrary real numbers follows from the Archimedean Property, for rationals or square roots of rationals, N can be constructed explicitly without invoking the completeness axiom.

(iii) Let x be any real number. Show that there is an integer $m \in \mathbb{Z}$ so that $m \leq x < m + 1$. Also, show that m is unique.

This integer m is called the integer part of x (or the floor function) and it is denoted by $\lfloor x \rfloor$.

(iv) Prove that for any real numbers x and y with $x < y$, there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

Problem 5. (10 points)

(i) Let S and T be nonempty bounded above subsets of $\mathbb{R}$. Define a subset $S + T$ as

$$S + T = \{s + t : s \in S \text{ and } t \in T\}.$$

Prove that $\sup(S + T) = \sup S + \sup T$.

(ii) Determine $\inf S$ and $\sup S$ for the set

$$S = \left\{ \frac{1}{p} : p \text{ is a prime number} \right\}.$$

Hint: You can use the fact that there are infinitely many prime numbers, so that, for each $n \in \mathbb{N}$, there is a prime number p such that $p \geq n$.

Problem 6. (10 points)

(i) Using the definition of the limit, prove that $\lim_{n \rightarrow \infty} \frac{4n^3 + 3n}{n^3 - 6} = 4$.

(ii) Let $\{a_n\}$ be a convergent sequence of integers. Show that there is $n_0 \in \mathbb{N}$ such that $a_n = a_{n_0}$ for all $n \geq n_0$.